

Contact

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Top Skills

Backend API Development

FastAPI

AI/ML Model Integration

Muhammad Ahmed afzal

AI Engineer | Building Scalable AI Agents & Automation Systems
| LangChain, FastAPI, Docker, GCP | Production-Ready Deep
Learning

Islāmābād, Pakistan

Summary

I'm an AI Engineer with two years of experience building LLM applications that actually get used in production, not just demoed once and shelved. I work mainly in FastAPI and Django, building backend systems for SaaS platforms, and spend most of my time on LLM orchestration with LangChain and LangGraph, including multi-agent setups. I build RAG pipelines (cosine similarity search, prompt tuning, retrieval optimization) and have automated workflows that used to require manual work from start to finish. I've also built real-time data visualization systems for AI applications running at scale. I work with models from OpenAI, Anthropic, ElevenLabs, and OpenRouter, and I care about keeping systems cheap to run and reliable in practice. Lately I'm spending more time thinking about LLM reliability, agent behavior, and evaluation frameworks, basically how to take research ideas and turn them into something that actually holds up in production.

Experience

iTechGemini Pvt Ltd

AI Engineer

September 2025 - Present (11 months)

Lahore, Punjab, Pakistan

As an AI Engineer at iTechGemini, I lead backend development for AI-powered mobile applications. I design, build, and deploy production-ready APIs that integrate machine learning models, automation workflows, and cloud-native services. My workload spans multiple APIs, each engineered for reliability, throughput, and real-world scalability.

I work across the full lifecycle model selection, integration, API architecture, optimisation, and GCP deployment. My solutions leverage models served through OpenRouter, Replicate, Hugging Face, and custom pipelines.

Depending on performance and cost targets, I deploy using App Engine, Cloud Run, or compute-optimised VM instances.

My focus is on shipping robust, low-latency AI services that plug directly into customer-facing applications, accelerating product delivery across the organisation.

Independent Contractor

AI Engineer – Independent Contractor

April 2025 - September 2025 (6 months)

Islāmābād, Pakistan

Drug Discovery AI: Enhanced 3DmolFormer (dual-channel transformer, 12 heads, 12 layers, 2 attention layers) with multi-objective reinforcement learning, enabling protein-specific docking predictions and generation of novel ligands for targeted drug discovery.

Luxury Booking Platform: Built SiteNest, a full-stack web application for premium apartment bookings using React, Node.js, PostgreSQL, and OpenAI, featuring real-time analytics, booking system, affiliate program, and AI-powered chatbot for customer assistance.

AI-Powered Mobile App: Developed PlayAround, a Flutter-based sports social app integrating real-time chat, venue booking, matchmaking, and skill tracking, deployed on Play Store with modern Material 3 UI/UX and smooth cross-platform performance.

Automotive Artificial Intelligence (AAI) GmbH

Artificial Intelligence Engineer

April 2022 - April 2025 (3 years 1 month)

Islāmābād, Pakistan

Engineered GoldenEYE automation system, replacing manual data dumping with network-based ingestion, cutting manual effort by ~85% and ensuring consistent hourly updates.

Designed and managed cloud-hosted MySQL databases on AWS (RDS/EC2), scaling to 100K+ records and supporting 10K+ daily transactions across TSR, Blockage, and QA modules.

Built real-time dashboards and automated Python pipelines, delivering <300ms latency insights, boosting operational efficiency by 70%, and enabling global access to analytics.

COMSATS University Islamabad

Research Assistant

September 2023 - January 2024 (5 months)

Islāmābād, Pakistan

Conducted advanced NLP research on sentiment analysis, text classification, and transformer models (BERT, GPT), improving classification accuracy by 15%.

Developed deep learning models using PyTorch and TensorFlow, including CNNs for image recognition, achieving 92% accuracy on benchmark datasets.

Engineered scalable model architectures for practical applications in healthcare imaging, emotion detection, and automated text analytics, bridging research with deployable AI solutions.

Education

COMSATS University Islamabad

Master's degree, Artificial Intelligence · (February 2024 - February 2026)

COMSATS University Islamabad

Bachelor's degree, Bioinformatics · (September 2019 - July 2023)